Archit Mangrulkar

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Education

Indian Institute of Technology Kharagpur **Nov 2020 – May 2024**
Bachelor of Technology in Computer Science *CGPA 8.57/10.00*

Work Experience

Samsung Research India - Bangalore **Jul 2024 – Present**
C++ Developer | Visual Intelligence Team *Bangalore, India*

- Commercialized Video Stabilization, QuickTake, Hyperlapse & Astrolapse on Samsung's flagship S series smart phones
- Mainlined camera pipelines for 60 fps video with **16.67 ms frame** window and support for 8K capture with 10-bit HDR
- Leveraged ARM-specific SIMD, OpenCL optimizations & supported gyro sensor listener sampling at **526 microseconds**

University of Alberta, Canada **May 2023 – Aug 2023**
Vision & Learning Lab | Multimodal Motion Capture *Edmonton, Canada*

- Developed large-scale multimodal dataset of humans in the wild of RGBD images, eye-tracking information, speech data
- Programmed a **multithreaded C++ server** and Python GUI client using **PyQt5** to achieve concurrent image capture
- Designed a 3D human pose tracking & clustering module for multi-view images with **YOLOv5**, DeepSORT, & DBSCAN

TU Dresden, Germany **May 2022 – Jul 2022**
Chair of Processor Design | Approximate Computing in Neural Networks *Dresden, Germany*

- Designed neural operators as C static libraries for fast **8-bit FPGA** inference of neural networks using TensorFlow Lite
- Explored cross-layer granularity combining signed 8-bit lookup tables, polynomial regression & EvoApproxLib multipliers
- Reduced inference time by **2x** using **OpenMP**'s multithreading, autovectorization, compiler flags & cache optimizations

Patents

Geometrically Consistent Multi-Camera Stabilization for XR Headsets (Patent Pending)

- Co-invented an AI driven multi-camera stabilization method enforcing geometric consistency for better visual experience
- Solved AR world-anchor loss during rapid, dynamic head motion through reinforcement learning inside stabilization loop
- Analyzed stabilization pipelines of **Apple, Meta**, and Xreal to pinpoint architectural limits & shape our novel claims

Key Projects

Order Matching Engine

- Built a limit order book & matching engine in C++ handling order entry, price-time priority matching, trade execution
- Achieved sub-microsecond latency (p99 **291.7 ns** over 10M orders) by removing heap allocation from the hot path
- Enabled safe concurrent order intake with a **lock-free queue**, stress-tested to guarantee zero lost or duplicate orders

Lock-Free Concurrency Toolkit

- Built **cache-optimized** C++ concurrency primitives: a multi-producer/multi-consumer queue & a concurrent hash map
- Achieved **2.14x** throughput over a mutex baseline under contention through cache-padding, **backoff** and prefetching
- Improved concurrent read throughput over a single global lock by sharding a hash map so readers never block each other

MyHTTP TCP Browser

- Designed a concurrent CLI browser for file retrieval from **HTTP/HTTPS** servers with **OpenSSL** for secure sockets
- Developed an HTTP 1.1 server handling GET/PUT for PDFs, TXTs, JPGs with AccessLog and status code responses

Publications

AAAI Conference on Artificial Intelligence 2026 (Under Review)

- Deployed multi-branch video stability model in C++/ONNX runtime achieveing a **15x speedup** over Python baseline
- Slashed **model size by 72%** via static INT8 quantization, solved Android Binder IPC limit & GPU memory contention
- Replaced RAFT model for optical flow with NeuFlow v2 demonstrating **44x FLOPS reduction** for edge deployment

Academic Achievements

- Workshop paper on Stance & Premise Detection accepted at International Conference on **Computational Linguistics**
- Offered department change to B.Tech CSE program for being among top 16 out of 1800 students securing 10.0 CGPA
- **MITACS Scholarship** 2023 - University of Alberta, Canada | **Erasmus+ Scholarship** 2022 - TU Dresden, Germany
- Secured **AIR 978** out of 150k candidates in JEE Advanced & **AIR 1517** out of 929k candidates in JEE Mains, 2020
- Attained the silver medal in **DevRev**'s Closed Domain QA Challenge at the **11th Inter IIT Tech Meet** 2023
- Won the gold medal in **DRDO**'s UAV Guided UGV Navigation Challenge at the **10th Inter IIT Tech Meet** 2022